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Segmental Prominence and the Modulated Carrier Signal

John Harris[1] and Larry M. Hyman[2]
[1]University College London
[2]University of California, Berkeley

Abstract

The segmental phonology of a wide range of West African languages shows clear contrastive asymmetries between different positions within the prosodic stem – a stem defined by a fixed number of syllables and various other restrictions. Stem-initial positions are strong in that they support maximal systems of consonant contrasts. Non-initial positions are weak not just because they support fewer contrasts but also because the consonants that typically occur there form a rather small set that can be intuitively described as weak. The notion of relative consonantal strength is hard to define using standard feature classifications. However, it can be captured by drawing on the model of speech as a modulated carrier signal: the stronger the consonant, the greater the extent to which it modulates the carrier. Consonants that are non-initial in the stem form a natural class in that they perturb the carrier to a relatively small extent. This definition of consonant strength provides us with a direct way of modelling how segmental effects, alongside prosodic effects, are fully integrated into the more general phenomenon of positional prominence.

Keywords: positional prominence, segmental prominence, consonant strength, modulated carrier signal

1. Introduction

Among the many insights and contributions to phonology that we owe to Eno-Abasi Urua, the most impactful on our own research is her extensive and ground-breaking work on the prosodic and segmental structure of Ibibio. In a number of works, Eno brings insightful clarity to the vowel and consonant systems and shows not only how they relate to syllable structure but also how they establish the central importance of the foot (Urua 1999, 2000, 2004, Akinlabi & Urua 2002, Harris & Urua 2001). Her extensive work continues to inspire contemporaneous and subsequent research by the above co-authors and other colleagues, including Akinlabi & Lee (2006), Harris (2004), Harris, Urua & Tang (2021), Hyman (2019: 64-66), Hyman *et al*. (2019: 196-7), as well as this chapter which we enthusiastically submit in her honor.

As will be seen further below, Eno's work on Ibibio represents a milestone in developing an understanding of how positional PROMINENCE in words is not just about stress accent but also has an important segmental aspect. The latter manifests itself most obviously as asymmetries in the contrastive potentials of different positions within domains such as the word, stem, or foot. Initial positions within these domains are 'strong' in the sense that they are able to support maximal systems of contrasts. Non-initial positions are weaker in the sense that contrasts are neutralized to varying extents. As will be seen below, this general pattern of positionally sensitive contrastive asymmetries is clearly in evidence not just in Ibibio but also in many other African languages.

The contrastive asymmetries typically go hand in hand with recurring phonetic asymmetries. The inventories of segments showing up in non-initial position are not only smaller (by definition, because of neutralization) but are also typically drawn from the same basic set, summarized in (1).

(1)

	Internal	Final
Laryngeal	Voiced	Voiceless
Manner	Plain stops	Unreleased stops (applosives)
	Nasals	Nasals
	Approximants	Approximants
Place	Labial-coronal-velar	Labial-coronal-velar
		Glottal stop, [h]
		Indeterminate nasal

Typically missing from non-initial inventories are aspirated stops, implosives, affricates, supralaryngeal fricatives, and place contrasts other than the canonical labial-coronal-velar series (apart from the palatal glide [y])—any of which might quite readily occur stem-initially in the same language.

Occurrence in the same set of environments is one of the criteria by which a set of sounds is deemed to form a natural class. However, how to define exactly what is natural about the class of non-initial sounds represented in (1) is perhaps not so obvious. We might appeal to some intuitive notion of segmental weakness or lack of prominence, but this is not easy to pin down using standard features or IPA classifications. However, the challenge can be met if we approach it from the perspective of the modulated carrier conception of speech (Dudley 1940, Traunmüller 1994, Ohala & Kawasaki-Fukumori 1997). As we explain below, this provides us with a phonetic definition of segmental strength: the stronger the segment, the greater the extent to which it modulates the carrier signal. The non-initial sounds in (1) are relatively weak in that they modulate the carrier signal to a diminished extent. Strong positions, by contrast, can host segments which modulate the carrier to a much greater degree—and which hence are capable in such positions of contrasting with less modulating segments.

This phonetic asymmetry is clearly evident in Ibibio, which we cite as our first example in §2. We follow this up with examples from Grassfields Bantu in §3 and from other African languages in §4. In §5, we discuss how the notion of segmental prominence can be fruitfully apprehended in terms of the modulated-carrier model of speech.

2. Ibibio

Positional asymmetries can be particularly stark in languages with PROSODIC STEMS – stems defined in terms of a fixed number of syllables and other restrictions (Hyman *et al*. 2019: 198, Downing 2006). The Nigeria-Cameroon border region is especially rich in languages of this type (Hyman 1990). It has been shown to be an areal characteristic of Northwest Bantu (Hyman 2008, Idiatov & Van de Velde 2016), and of genetically diverse languages throughout Sub-Saharan Africa (Lionnet 2017, Hyman *et al*. 2019). As Urua (2000) has amply documented, Ibibio is one such language, and provides a good starting point for several reasons. First, it provides us with a beautiful case of a prosodic stem with contrastive asymmetries. In this case, the stem is subject to a limit of two syllables (root + possible suffix). As described below, the contrastive potential of the two syllables is strongly skewed in favour of the first, suggesting that the stem is equivalent to a trochaic foot. (In §3 and §4 we will see that other languages allow for trisyllabic and quadrisyllabic prosodic stems.) Second, unlike widespread cases of onset (C1) versus coda (C2) asymmetries, Ibibio places constraints on the second stem consonant not only when it is a coda but also when it is the onset of the second syllable. Finally, Urua and others have provided detailed documentation, description, and theoretical discussion of the Ibibio facts (Urua 2000, Harris & Urua 2001, and Akinlabi & Urua 2002).

As a starting point, consider in (2) the striking consonant distributions, where CC stands for a geminate, [y] symbolizes a palatal glide, and the coda consonants are shown as realized before pause.

(2)	a.	prosodic stem structures:	CV, CVC, CVVC, CVCV, CVVCV, CVCCV												
	b.	stem-initial consonants:	b	f	m	t̪	d	s	n	y	ɲ	k	ŋ	kp	w
	c.	coda consonants:	p		m	t̪			n	y		k	ŋ		
	d.	intervocalic VCV:	β		m	ɾ			n			ɣ	ŋ		
	e.	intervocalic VCCV:	pp		mm	t̪t̪			nn	yy		kk	ŋŋ		

Two aspects of the contrastive asymmetries evident in this display strike us right away. First, there are more underlying consonant contrasts stem-initially than non-initially. Of particular note are the manner and voicing neutralizations of the C1 contrasts /b/-/f/ and /t̪/-/d/-/s/. Second, non-initial stops are realized as applosives in coda position but weaken to approximants [β, ɾ, ɣ] intervocalically both stem-internally and when word-final followed by a vowel-initial word.[1] This can be seen in alternations such as those in (3) (Urua 1999: 247):

(3)		*root*		*passive/reflexive*	
	/b/	díp	'hide'	díβé	'hide oneself'
	/t/	yàt	'wear (on head)'	yàɾá	'wear on one's head'
	/k/	kɔ́ɔ́k	'stack	kɔ́ɣɔ́	'be stacked'

Stem-initial stops resist weakening even when intervocalic, i.e. when preceded by a vowel prefix.

As shown in (2a), the maximal size of the stem in Ibibio is CVVCV or CVCCV, a heavy-light bisyllabic template that is obligatory in certain constructions. The prosodic shape of the stem clearly indicates that both the distribution and realization of segmental contrasts in Ibibio are foot-based (Urua 1999: 247, 253). A good illustration of this is to be found in the behavior of the negative suffix /-ke/. The /k/ weakens when the suffix forms the weak syllable of a foot, as in (4a), but not when it falls outside the foot, as in (4b):

(4)						
	a.	dí	→	dííɣé	'not come'	/díV-ké/
		dó	→	dóóɣó	'not be'	/dóV-ké/
	b.	tòβó	→	tòβóké	'not make an order'	/tòpó-ké/
		sàŋá	→	sàŋáké	'not walk'	/sàŋá-ké/

Based on both segmental and prosodic facts, similar conclusions about foot-sized stems can be drawn from other Cross-River languages spoken in southeastern Nigeria, e.g. Gokana (Hyman 2011), Kana (Ikoro 1996), and Leggbó (Hyman & Udoh 2007).

3. Grassfields Bantu

Crossing the border into Cameroon, we find a similar type of distribution within the maximally bisyllabic foot in the Fe'fe' dialect of Bamileke, a Grassfields Bantu (Bantoid) language (Hyman 1972):

[1]The phonetic symbols in (2c) suggest that the stops resulting from the neutralization of laryngeal contrasts in codas line up with the initial voiceless series in (2b). However, the unreleased realization of the coda archiphonemes makes it difficult to identify them with either the voiced or the voiceless initials. Urua notes that speakers may hesitate to make a choice: 'In word final position /b/ is realized as unreleased, and may be perceived as a [p-]. Even for native Ibibio speakers, it is difficult to distinguish between [b] and [p] in word-final position' (2000: 41-2). We have noted the same hesitation among speakers of other West African languages, e.g. Bamileke-Fe'fe' (to be discussed in §3), and also Tibeto-Burman languages lacking a voicing contrast in word-final unreleased stops.

(5)	a.	prosodic stem structures:	CV, CVC, CVV, CVCV, CVVCV												
	b.	initial consonants:	p	f		t	s		č	k			ʔ	h	
				v	m	l	z	n	y	ɣ	ŋ	w			
	c.	intervocalic VCV:	b~β	m		l		n		ɣ			ʔ	h	
	d.	coda consonants:	p		m	t		n		k			ʔ	h	

As seen, there are 17 contrastive C1 stem-initial consonants versus seven C2 intervocalic and coda consonants. Like Ibibio, coda consonants are either nasal or oral stops; the latter are unreleased before a pause but become voiced and continuant before a vowel. (See note 1, however.) However, Fe'fe' differs from Ibibio in two ways. First, initially the labial stop is [p] rather than [b]. While many languages are like Ibibio in lacking [p] in onset, if not all positions, Fe'fe' definitely favors initial [p], restricting initial [b] to exactly four words. Three of these are grammatical morphemes: demonstratives [béè] 'this/these', [báa] 'that/those (near hearer)', [bíi] 'those' (far from speaker and hearer", plus [bɑ] 'be', which can also be pronounced [pɑ] (Hyman 1972: 62). The second difference is that proto *d, *ǰ, *g and *gw are pronounced [l], [y], [ɣ] and [w] even in C1 position. These consonants harden, however, after a nasal prefix, thus producing a nasal + voiced stop complex in the following inflected verb forms (p. 46):

(6)	a.	N+l → nd	lɛn / ndɛn	'to say'
	b.	N+y → nǰ	yɛèn / nǰén	'to cork'
	c.	N+ ɣ → ŋg	ɣèn / ŋgén	'to go'
	d.	N+w → ŋgw	wɛn / ŋgwɛn	'to request aid'
	e.	N+p → mb	pɛn / mbɛn	'to accept'

(6e) shows that [p] becomes voiced in the same derived context. If one were to compare the treatment of the labial stop in Ibibio and Fe'fe', it would seem that they have gone in opposite directions: in initial position, Ibibio disfavors [p], while Fe'fe' favors it. This may seem surprising, given that Fe'fe' otherwise lenites C1 voiced consonants. What the two languages have in common, however, is that there is no laryngeal contrast in bilabial stops.

Stem-based contrastive asymmetries of the pattern just described for Bamileke Fe'fe' can be seen right across Grasslands Bantu. Table 1 summarizes the diminished coda inventories found in a sample of these languages. Like Bamileke Fe'fe', all of the languages show neutralization of laryngeal contrasts in this position, and many show neutralization of manner (e.g. in the absence of supralaryngeal fricatives). Some also show partial or total neutralization of place.

Stem-internally, all of these languages show special intervocalic realizations of the stop consonants which, along with many other indications, confirm that this position too is segmentally weak.

Table 1. Coda consonants in Grassfields Bantu (Hyman et al 2019:193)

Language	Subgroup	P	T	K	ʔ	m	n	ɲ	ŋ	f	s	h	l	r	w	y	Totals
Medumba	Eastern	p	t	k	ʔ	m	n		ŋ								7
Fe'fe'	Eastern	p	t	k	ʔ	m	n					h					7
Yemba	Eastern	p	t	k	ʔ	m			ŋ								6
Bamena	Eastern	p	t	k	ʔ	m						h					6
Ghomala	Eastern	p		k	ʔ	m			ŋ								5
Bamunkumbit	Eastern	p	t			m	n										4
Shingu	Eastern	p	t	k	ʔ							h					5
Ba'angu Akum	Eastern					m	n		ŋ								3
Limbum	Eastern	b			ʔ				ŋ					r			4
Mfumte	Eastern				ʔ												1
Moghamo	Momo	p	t	k	ʔ	m	n		ŋ								7
Mundani	Momo	p	t			m	n		ŋ								5
Ambele	Momo	p	t	k		m			ŋ								5
Weh	Ring	p	t		ʔ	m	n		ŋ						w	y	8
Bum	Ring		t	k	ʔ	m	n		ŋ	f	s	h					9
Oku	Ring			k		m	n		ŋ	f	s		l				7
Kom	Ring				ʔ	m	n		ŋ	f	s		l				7
Babanki	Ring				ʔ	m	n		ŋ	f	s						6
Lamnso	Ring					m	n		ŋ	v				r		y	6
Babungo	Ring				ʔ				ŋ								2
Bamessing	Ring								ŋ			h			w	y	4

4. Other languages

Once we venture a bit more east into Northwest Bantu, the situation is similar, but expanded by the fact that the prosodic stem can be larger than a bisyllabic foot. Still, we see the same kind of distributional asymmetries contrasting successive consonant positions. A case in point is Koyo, a Bantu language spoken in République du Congo, where codas are disallowed and the stem can have up to four syllables (Hyman 2008:330-1):

(7) a. stem structures: CV, CVV, CVCV, CVCVCV

b.

C1:	p	b	w	m	mb	t	l	s	n	nd	ts	dz	y	ɲ	ndz	k	ŋg	h	
C2:		b		m	mb	r	l	s	n	nd			y	ɲ	ndz	g			
C3:				m		r	l	s	n							g			
C4:																g			

In this case, where there are potentially four consonants in the stem, we observe progressively fewer contrasts in C2 versus C3 versus C4 position, the last of which can only be filled by the verbal durative suffix /-gV/. Once again, voiceless stops are found only in stem-initial position. The examples in (8) show that /t/ and /k/ are realized are [r] and [g] internally, and are not affected stem-initially even if preceded by a vowel prefix:

(8) a. /i-tɔtɔ/ → i-tɔrɔ 'banana'
b. /o-tokiti/ → o-togiri 'sweat'

Other Northwest Bantu languages with distributional asymmetries similar to Koyo include Basaa (Lemb & Degastines 1973, Hyman 1990, Bitjaa-Kody 1990, a.o.), Tiene (Ellington 1977, Hyman 2010), Eton (Van de Velde 2008, Idiatov & Van de Velde 2016),

Kukuya (Paulian 1974, Hyman 1987), Punu (Fontanay 1980, Kwenzi-Mikala 1980, Hyman 2008),Yaka (Hyman 1998), and many more (see Lionnet 2017 and Hyman *et al*. 2019).

Some of these languages further illustrate how the prosodic stem can be larger than a two-syllable foot. In addition to Koyo, we find three-syllable stems in for example Basaa, Tiene, and Kukuya, while four-syllable stems occur in Punu and Yaka. The Tiene and Kukuya cases are interesting because the maximum CVCVCV prosodic stem not only restricts which consonants can occur as C2 and C3, but also which C2-C3 orders are OK. For instance, in a Tiene CVCVCV stem, C2 must be coronal, and C3 must be labial or velar (see Hyman 2010 for attempts to understand why this might be the case).

Our focus here has been on the languages of West Africa, but similar examples of stem-based contrastive asymmetries are to be found right across the world's languages. One rich source of examples is provided by the Khoesan languages of southern Africa, where extensive inventories of click consonants and their various accompaniments help make the stem-based asymmetries particularly dramatic. For example, in Ju|'hoansi the prosodic stem is CV(V)CV, where C1 supports 88 distinct consonants and C2 only four, two nasals and two approximants (Miller 2010). In !Xóõ, the stem is CVC, CVV, or CVCV, with C1 hosting 113 consonants and C2 hosting just two, both nasals (Traill 1985). Even further afield, Tibeto-Burman furnishes us with many examples of languages with coda inventories largely restricted to unreleased stops, nasals, and approximants (Namkung 1996).

5. Segmental prominence and the modulated carrier signal

Based on our review of stem-based distributional asymmetries in different languages, we find support for the set of consonants that preferentially appear in C2 position presented in (1), repeated in (9).

(9)

	Internal	Final
Laryngeal	Voiced	Voiceless
Manner	Plain stops	Applosives
	Nasals	Nasals
	Approximants	Approximants
Place	Labial-coronal-velar	Labial-coronal-velar
		Glottal stop, [h]
		Indeterminate nasal

What, if anything, unifies the class of consonants in (9), apart from their occurrence in the same general environment? An answer to this question can be found in the conception of speech as a modulated carrier signal (Dudley 1940, Traunmüller 1994, Ohala & Kawasaki-Fukumori 1997).

The carrier signal in speech can be thought of as the schwa-like sound that is produced by a neutrally open vocal tract. Phonological information in speech is conveyed by the varying ways in which the carrier is modulated along the auditory-acoustic dimensions of amplitude, periodicity, spectral shape, fundamental frequency, and time. The magnitude of the modulation created by a given consonant can be measured in terms of the distance it travels through this multidimensional space as it moves away from the acoustic baseline set by the carrier (Ohala & Kawasaki-Fukumori 1997). The longer the distance, the larger the modulation.

The modulated-carrier model has been invoked as a way of fleshing out the notion of segmental strength: stronger segments modulate the carrier to a greater extent than weaker segments (Harris & Urua 2001, Harris 2004, Harris, Urua & Tang 2021). In this way, all types of consonant lenition can be defined as diminishing modulations. A related notion is to be found in the work of Kingston (2008) and Katz (2016), who characterize certain types of

lenition as reducing the extent to which a consonant interrupts the stream of speech. The latter conception applies only to spirantization/vocalization, such as occurs in Spanish for example. This primarily affects the amplitude of a consonant, the dimension most relevant to one of the consonant types in (9), namely approximants. A modulated-carrier approach to segmental strength is more encompassing than this, since it covers all forms of lenition, including those that affect dimensions other than amplitude. For example, debuccalization, which is typically the process behind the glottal consonants in (9), affects spectral shape. Loss or suppression of plosive release, the effect that gives us the applosives in (9), affects periodicity.

It is this multidimensional aspect of the modulated-carrier model that provides us with a unitary definition of the class of non-initial segments in (9). Applying it to the distributional asymmetries we have reviewed here yields pairwise comparisons between different types of consonant such as those listed in (10).

(10)

LARGER MODULATION	SMALLER MODULATION
Plosive	Applosive
Plosive with laryngeal accompaniment	Plain stop
Intervocalic voiceless stop	Intervocalic voiced stop
Plosive	Nasal
Plosive	Glottal stop
Supralaryngeal fricative	Glottal fricative
Non-approximant consonant	Approximant

The consonant types with smaller modulations are those that preferentially show up in non-initial positions. In most of the languages reviewed here, consonants with larger modulations are restricted to initial position. That is not to say that the less-modulating consonants are necessarily excluded from initial position; see for example the Fe'fe' distributions in (5). This simply reflects the fact that initial position sustains more contrasts than non-initial positions. Nevertheless, it is true that certain types of less-modulating consonants cannot appear initially. Applosives, for example, are by definition restricted to coda position.

These distributional asymmetries can only be partially explained by differences in the relative robustness of auditory-acoustic cues to consonants in different local environments. The immediately local environment of prevocalic position is known to be the most favorable context for cueing the identity of a consonant, which plausibly explains why codas disproportionately host neutralization (Wright 2004, Henke *et al*. 2012). However, in the languages we have described here, the prevocalic environment is also present stem-medially, as well as stem-finally when a vowel-initial morpheme follows; yet neither of these positions supports the range of consonant contrasts found in initial position. Clearly then, the stem is playing an independent role in regulating the distributional asymmetries.

The comparisons in (10) illustrate how the modulated-carrier account of segmental strength applies to manner and laryngeal contrasts. As for place, the comparisons also illustrate how the account supports a distinction between more-modulating, supralaryngeal consonants and less-modulating, glottal (placeless) consonants. Is there any evidence of similar asymmetries among different supralaryngeal places? Previous attempts to establish universal strength relations based on place were not particularly successful (see Smith 1981 for a summary). However, our review of the languages above throws up at least one place-based pattern that invites analysis according to a modulated-carrier account of segmental strength.

A glance at the Ibibio consonant distributions in (2) suggests a place-based strength asymmetry within the class of labial stops. Note the complementary distribution of [kp] and [p]: the former only occurs in strong, initial position, while the latter only occurs in weak, coda

position. The defective distribution of [p] in Ibibio is a positionally limited version of the broader pattern found in many languages where [p] is absent from the stop system altogether. The [p]-gap is plausibly explained by bilabials having the least robust release-burst cue of all the plain stops, a result of there being no constriction downstream from the labial closure that would otherwise increase the intensity of the burst (Ohala 2005). Doubly articulated [kp] has a more intense release burst than plain [p], due to there being a labial constriction downstream from the dorsal closure (Connell 1994). In other words, [kp] modulates the carrier to a greater extent than [p]. This is consistent with the asymmetric distribution of these two sounds within the Ibibio stem. We can think of [kp] as an acoustically enhanced version of an otherwise vulnerable [p], parallel to the way prenasalization or implosion enhance otherwise vulnerable voicing in stops (Ohala 2018).

This strength account of labials stops is bolstered by the fact that similar distributional asymmetries involving these sounds are found in some of the other languages reviewed here.

First, several of the Grassfields Bantu languages are like Ibibio in lacking a distinction between [b] and [p] in initial position. In most cases, as in Ibibio, the contrast is neutralized in favor of [b], e.g., Kom (Shultz 1997), Babanki (Faytak & Akumbu 2020), and Bamessing (DeVries 2008). The only exception, as noted in §3, is Fe'fe', which favors initial [p] – though even here a voiced allophone crops up under post-nasal voicing.

Second, again like Ibibio, several languages restrict labial-velars to initial position. Note the total absence of labial-velars from the display of codas in Grassfields Bantu in Table 1. In Weh, we find [kp, gb] initially but plain labial stops elsewhere (Davison 2009). Oku has initial [kw, gw] (contrasting with [b, k, g]), while the only coda stop is [b] (Davis 1992). In Lamnso, [kp, gb] occur initially, while the only coda stop is glottal (McGarrity & Botne 2001).

Besides supporting a definition of what is natural about the class of non-initial consonants, a modulated-carrier definition of segmental strength provides us with a direct way of modelling how segmental effects, alongside prosodic effects, are fully integrated into the more general phenomenon of positional prominence. It helps deepen our understanding of how segmental phonology, like stress accent, can play a communicatively valuable role in signaling word junctures in speech. Indeed, since all of the data we have surveyed here come from languages that have been analyzed exclusively with tone, we can see very clearly how segmental prominence can manifest itself even in the absence of stress accent. Modulation-reducing processes that target non-initial consonants but leave initial consonants unscathed help accentuate the auditory-acoustic difference between the two positions. This difference provides the listener with potentially valuable demarcative cues to prosodic and morphological domain structure.

6. Conclusion

In conclusion, we note that none of the above generalizations would have been possible without the careful description and analysis of pioneers such as Eno Urua. We note that she has moved on to other cutting-edge investigations which are yielding interesting results likely to influence future generations as well. We congratulate her, thank her, and wish her well in these endeavors too.

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